PHISHING WEBSITES DETECTION USING MACHINE LEARNING

A project report submitted in partial fulfillment for the award of the degree of

BACHELOR OF TECHNOLOGY

IN

COMPUTER SCIENCE AND ENGINEERING

by

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CERTIFICATE

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under my guidance and supervision for the partial fulfillment of the requirements for the award of B.Tech degree in **COMPUTER SCIENCE AND ENGINEERING.** This project is the result of their own effort and that it has not been submitted to any other University or Institution for the award of any degree or diploma other than specified above

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ABSTRACT

The growing threat of phishing attacks, which exploit the anonymity of the internet to trick users into giving away sensitive information. Phishing websites typically mimic legitimate websites closely, both in design and in their URLs, making them difficult for ordinary users to detect. Phishing attacks are commonly carried out via email links or social media, with the phishing website's goal being to collect personal information or install malware on the users Devices.

Phishing is a widespread tactic used to trick gullible people into disclosing their personal information by using bogus websites. Phishing website URLs are designed to steal personal data, including user names, passwords, and online financial activities. Phishers employ websites that resemble those genuine websites both aesthetically and linguistically. Utilizing anti-phishing methods to identify phishing is necessary to stop the rapid advancement of phishing techniques as a result of advancing technology. A strong tool for thwarting phishing assaults is machine learning.

In response to these limitations, The proposed system architecture employs a wrapper feature selection technique to enhance the accuracy of the machine learning classifiers. The selection of Features (domain-based attributes, URL-based attributes) allows the model to detect a wide variety of phishing indicators, even in URLs that mimic legitimate ones.

Keywords: Selection Features, Machine Learning, Random Forest, Predictive Analytics, Model Evaluation, Gradient Boosting Classification.

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LIST OF ABBREVIATIONS

IDE	Integrated Development Environment
PWD	Phishing Websites Detection
CSV	Comma-separated values
ML	Machine Learning
UML	Unified Modeling Language

Chapter 1

INTRODUCTION

As digital interactions grow, so does the threat of cyberattacks. Among these, phishing remains one of the most dangerous, as it targets users by tricking them into sharing sensitive information on fake websites. Phishing websites often appear nearly identical to legitimate ones, using subtle differences to deceive users. According to recent statistics, over 80% of all reported cyberattacks involve phishing, indicating the severity and impact of the problem. As attackers continuously enhance their methods, including using HTTPS and social engineering, it has become crucial to develop more robust, adaptable systems.

Traditional phishing detection systems rely on blacklist-based or rule-based techniques, which often fail to keep up with the ever-evolving tactics employed by cybercriminals. As phishing techniques become more sophisticated, there is a growing need for intelligent and adaptive solutions. In response to this challenge, machine learning (ML) has emerged as a powerful tool for automating and enhancing phishing detection. ML models can learn from historical data and effectively distinguish between legitimate and malicious URLs based on features such as domain characteristics, URL structure, and protocol usage.

This project proposes a machine learning-based phishing URL detection system that leverages multiple algorithms—including Random Forest, Decision Tree, SVM, and KNN—to analyze and classify URLs. The system is trained on a comprehensive dataset obtained from Kaggle, consisting of both phishing and legitimate URLs. After thorough feature extraction and preprocessing, the models are evaluated for their performance. Among them, Random Forest achieved the highest accuracy while maintaining computational efficiency, making it ideal for deployment even on low- resource devices.

Chapter 2
PROJECT DESCRIPTION

2.1 PROBLEM DEFINITION

With the increasing reliance on digital platforms for communication, banking, and e- commerce, phishing attacks have become a major cybersecurity concern. These attacks exploit human trust by using deceptive websites that mimic legitimate ones to trick users into revealing sensitive information. Traditional methods of phishing detection, such as maintaining blacklists or using manually crafted rules, are limited in their ability to detect newly emerging or obfuscated phishing attempts. These static approaches often fail to adapt to the dynamic and ever-changing nature of phishing techniques.

The main challenge lies in accurately and efficiently identifying malicious URLs before users fall victim to them. Furthermore, most advanced security systems demand significant computational resources, making them unsuitable for devices with limited hardware capabilities. Therefore, there is a critical need for an intelligent, adaptable, and resource-efficient solution that can automatically classify URLs as legitimate or phishing using minimal human intervention. The solution should leverage machine learning techniques that can learn patterns from existing data and generalize effectively to new, unseen threats.

This project aims to address these challenges by developing a machine learning-based phishing detection system that can accurately identify phishing URLs using features such as URL length, domain age, presence of special characters, and HTTPS status. The model should be lightweight enough to run on basic systems while maintaining high detection accuracy. The Ultimate goal of this project is to improve the accuracy and efficiency of detection of Phishing websites.

2.2 PROJECT DETAILS

Data:

The Phishing websites Data is collected from the online sources such as Kaggle and other sites in the csv format. Collected data is processed to remove any missing values, noises, duplicates in the data set. After the collection of the dataset, we split the dataset into Training Data and Testing Data. In this project the Data set is split as 80% and 20%. The 80% is used for training the prediction model and 20% is used for testing the prediction model.

Prediction System:

Gradient Boosting Classification Machine Learning Algorithm is used to train the prediction model which is more accurate. It is used for predicting the categorical. This Algorithm predicts the output of a categorical dependent variable. It can be either Yes or No, 0 or 1, true or False, etc. The metric used for evaluating the model in the project is accuracy score. Accuracy score is the percentage of correctly predicted instances in the data set.

The Prediction System is built using the Trained and Evaluated model. Building process of the Prediction System includes several steps. The Prediction System takes parameters from the user and predicts whether the Website is Legitimate(safe) or Phishing(unsafe).

User Interface:

A flexible User Interface is developed for the use of Phishing websties Prediction System. This user interface is adaptable and User friendly, JavaScript and Flask an open-source framework for Machine learning is used to create the user interface of Prediction of phishing websites using Machine Learning. The User Interface created for Prediction of Heart Disease using Machine Learning is a webpage which takes the parameters from the user and sends to the prediction system and the result is displayed to the user. From this the user can predict the result.

Chapter 3

COMPUTATIONAL ENVIRONMENT

3.1 SOFTWARE REQUIREMENTS

Platform	: Windows 10
Programming Environment (IDE)	: Jupyter Notebook
Design	: Star UML
Coding Language	: Python
Frontend	: Java Script

3.2 HARDWARE REQUIREMENTS

Processor	: Inter i3, AMD Ryzen3 or higher
Speed	: 1.6 ghz or higher
RAM	: 4 GB or higher
Storage Disk	: 500 GB or higher

3.3 FUNCTIONAL REQUIREMENTS

Phishing Websites Dataset:

The Phishing Websites dataset consists of the various factors related to heart which helps in prediction of heart disease in that individual. The parameters or the features or factors in the heart disease dataset are as follows;

UsingIP: (categorical - signed numeric) : { -1,1 }
LongURL: (categorical - signed numeric) : { 1,0,-1 }
ShortURL: (categorical - signed numeric) : { 1,-1 }
Symbol@: (categorical - signed numeric) : { 1,-1 }

Redirecting:// (categorical - signed numeric) : { -1,1 }
PrefixSuffix-: (categorical - signed numeric) : { -1,1 }
SubDomains: (categorical - signed numeric) : { -1,0,1 }
HTTPS: (categorical - signed numeric) : { -1,1,0 }

LinksInScriptTags: (categorical - signed numeric) : { -1,0,1 }

ServerFormHandler: (categorical - signed numeric) : { -1,0,1 }

InfoEmail: (categorical - signed numeric) : { -1,1 }

AbnormalURL: (categorical - signed numeric) : { -1,1 }

WebsiteForwarding: (categorical - signed numeric) : { 0,1 }

StatusBarCust: (categorical - signed numeric) : { -1,1 }

DisableRightClick: (categorical - signed numeric) : { -1,1 }

UsingPopupWindow: (categorical - signed numeric) : { -1,1 }

3.4 : SOFTWARE FEATURES

Python:

The most abundantly used general level programming language. It is used for both a small scale and big scale systems. It can easily be interpreted. It is said to support multiple programming paradigms. It includes features of procedural, object-oriented, and functional programming together. It is already garbage-collected which makes it more efficient.

It is designed to be easy to read, write and understand, and it is popular for its simplicity, versatility and strong support for various programming paradigms such as procedural, object-oriented and functional programming. Some of the main features of python are as follows; Readability, Portability, Libraries and frameworks, Interpreted. Python is used in a wide range of applications, such as web development, data analysis, scientific computing, machine learning, and artificial intelligence. Its versatility, ease of use, and robust community support make it a popular choice for developers and businesses alike.

Advantages:

1. Simple and easy to learn: Python has a simple and easy-to-understand syntax that makes it easy for beginners to learn. Its code is readable and intuitive, which makes it easy to write

and maintain.

2. Large standard library: Python has a large and comprehensive standard library that provides a wide range of modules and functions, including tools for data processing, network programming, web development, and more. This saves time for developers and reduces the amount of code they need to write.
3. Cross-platform compatibility: Python is a platform-independent language, which means that Python code can run on any platform, including Windows, macOS, Linux, and UNIX.
4. Rapid prototyping: Python's simplicity and ease of use make it an ideal language for rapid prototyping. It allows developers to quickly write and test code without worrying about the complexities of other languages.

5. High-level language: Python is a high-level language that allows developers to write complex programs using fewer lines of code. This saves time and makes the code easier to read and maintain

NumPy:

NumPy is a Python library that provides support for large, multi-dimensional arrays and matrices, along with a large collection of high-level mathematical functions to operate on these arrays. NumPy stands for "Numerical Python" and is widely used for scientific computing, data analysis, and machine learning. This basically helps us deal with large datasets, matrices, and multi- dimensional arrays.

Pandas:

Pandas is an open-source data analysis and manipulation library for Python. It provides a flexible and high-level interface for working with tabular data in a variety of formats, such as CSV, Excel, SQL databases, and more. It is a library that is written in python language. It helps us with the analyzing of data. It also provides us with tools and functions to manipulate a large amount of data.

Sklearn:

Scikit-learn, also known as sklearn, is a popular Python library for machine learning. It provides a wide range of algorithms and tools for data preprocessing, model selection,

feature extraction, and more. It is a library used in machine learning in python programming language. It mainly helps us with the classification of data, regression of models, and in clustering algorithms. These algorithms include SVM, logistic regression etc.

Flask Framework:

Flask is a web application framework written in Python. Armin Ronacher, who leads an international group of Python enthusiasts named Pocco, develops it. Flask is based on Werkzeug WSGI toolkit and Jinja2 template engine. Both are Pocco projects.

Http protocol is the foundation of data communication in world wide web. Different methods of data retrieval from specified URL are defined in this protocol.

By default, the Flask route responds to the GET requests. However, this preference can be altered by providing methods argument to route() decorator.

In order to demonstrate the use of POST method in URL routing, first let us create an HTML form and use the POST method to send form data to a URL.

Juypter NoteBook:

Jupyter Notebook is an open-source web application that allows you to create and share documents that contain Live code, Equations, Visualizations, Text. It is especially popular in data science, machine learning, and academic research because it makes it easy to write and test code in small sections called cells. Jupyter supports multiple programming languages, with Python being the most commonly used. Users can also include rich text using Markdown, making it ideal for writing explanations, documentation, and reports alongside the code. Overall, Jupyter Notebook is a powerful and interactive platform for coding, learning, and presenting data-driven projects.

Chapter 4

FEASIBILITY STUDY

A feasibility study is the foundation for any successful project. It serves as the initial design phase, where different aspects of the proposed system are examined to determine whether the project is viable, practical, and worthwhile. This stage brings together technical knowledge, resource planning, time estimation, and cost analysis to form a comprehensive overview of what the project entails and whether it should move forward.

The purpose of conducting a feasibility study is to identify potential challenges early in the process and to ensure that the project has a high likelihood of success. It begins by estimating the level of technical expertise required, identifying who will be responsible for various tasks, and assessing both quantitative (like budget, manpower, infrastructure) and qualitative (like user satisfaction, ease of use, and stakeholder acceptance) aspects of the project. Additionally, it outlines the key milestones, proposes a realistic timeline, and gives an early estimate of the cost involved in completing the project.

As part of the system analysis phase, feasibility is further evaluated to ensure that the proposed solution is not only technically achievable but also aligns with organizational goals and constraints. It involves a deeper understanding of system requirements and examines whether the available resources—such as hardware, software, personnel, and budget—are sufficient to implement the system. The cost-benefit analysis is also performed here to compare the potential advantages of the system against the expected costs.

The main goal of this process is to analyze multiple solution alternatives, assess their feasibility in practical terms, and select the one that is most effective and efficient. The final output of the feasibility study is a business proposal that outlines the best-suited solution along with a rough implementation plan, budget, and benefit analysis. Ultimately,

this step is critical in reducing risks, avoiding unnecessary expenditure, and guiding decision-makers to invest in projects that are realistically achievable and beneficial to the organization.

4.1 TECHNICAL FEASIBILITY

Technical feasibility focuses on evaluating whether the current technological resources, tools, and expertise are sufficient to successfully develop and implement the Multi-Disease Prediction System within the allocated time frame and budget. It plays a crucial role in determining whether the system requirements, including complex machine learning operations, data integration, and real-time predictions, can be supported by the existing infrastructure and tools available.

For this project, technical feasibility begins with an assessment of the hardware and software capabilities required to handle tasks such as data preprocessing, model training, testing, and deployment. Additionally, the technical feasibility study considers the technical skills and experience of the development team. It ensures that the team has sufficient knowledge in Python programming, machine learning concepts, front-end development using Streamlit and CSS, and data handling using databases like SQLite. If there are any gaps, the feasibility study recommends training sessions or bringing in additional technical experts to bridge them.

Another important aspect is evaluating the reliability and stability of the technologies being used. For instance, tools like Python, Scikit-learn, and Streamlit are widely used, well-documented, and supported by large communities. This ensures that if any issues arise during development or deployment, the team can easily access forums, resources, or even third-party developers for help or suggestions.

4.2 OPERATIONAL FEASIBILITY

Operational feasibility evaluates whether the Multi-Disease Prediction System will function effectively in the real-world environment, meeting user needs and solving

healthcare-related problems. It focuses on how well the system aligns with user expectations, how easily it can be adopted, and how smoothly it can operate after deployment.

- **Identifying Key User Needs:** The system addresses a high-priority requirement early and accurate disease prediction—which is crucial for timely healthcare decisions.
- **Solution Acceptance:** The proposed solution, built using machine learning models and a user-friendly Streamlit interface, is practical and acceptable for medical staff and health practitioners.
- **User Adaptability:** Since the interface is designed to be simple and interactive, users with basic computer knowledge can quickly adapt without extensive training.
- **Organizational Fit:** The system aligns well with the goals of healthcare institutions aiming to digitize diagnosis and improve patient care efficiency.

4.3 ECONOMIC FEASIBILITY

Economic feasibility evaluates whether the Multi-Disease Prediction System is a financially viable project that can deliver long-term benefits to the healthcare sector. It involves analyzing development costs, resource investments, and potential returns in terms of efficiency, accuracy, and improved medical decision-making.

- **Development Cost vs Long-Term Gains:** Although initial costs are involved in developing the system (ML model training, software setup, data collection), it can significantly reduce manual diagnostic efforts, thereby saving time and cost in the long run.
- **Resource Investment:** Expenses include hardware upgrades, development tools (like Python, Streamlit), and database systems (SQLite).

Chapter 5

SYSTEM ANALYSIS

5.1 EXISITNG SYSTEM

- Phishing detection has traditionally relied on approaches like URL blacklisting and heuristic-based detection. Blacklisting involves maintaining a database of known phishing URLs, which is referenced every time a URL is visited. Cybercriminals can easily bypass blacklists by frequently changing URLs, and rule-based systems also cannot anticipate all variations. This approach, while straightforward, requires frequent updates, as new phishing sites emerge constantly.
- In the existing system they have used Logistic Regression, Multinomial Naive Bayes, and XG Boost are the machine learning methods that are compared. The Logistic Regression algorithm outperforms the other two. The model is preprocessed in the proposed system, the words are tokenized, and stemming is performed.

Drawbacks of Existing System

- The existing models have low latency.
- Existing systems do not have a specific user interface.
- The existing system model fails to predict a continuous outcome. It only works when the dependent or outcome variable is dichotomous.
- The existing system model may not be accurate if the sample size is too small.
- The existing may lead to overfitting problem.

5.2 PROPOSED SYSTEM

- We have developed our project using a website as a platform for all the users. This is an interactive and responsive website that will be used to detect whether a website is legitimate or phishing. This website is made using different web designing languages which include HTML, CSS, Javascript and Flask framework in Python. The basic structure of the website is made with the help of HTML. CSS is used to add effects to the website and make it more attractive and user-friendly. It must be noted that the website is created for all users, hence it must be easy to operate with and no user should face any difficulty while making its use.
- The proposed system is trained with the dataset consists of different features and note that the dataset don't contain any website URL. The dataset consists of different features that are to be taken into consideration while determining a website URL as legitimate or phishing. The proposed system is developed using the Gradient Boosting Classifier.
- After the system is trained with the dataset, the classifier identifies the given URL dependent on the preparation information that is if the site is phishing it prompts the user that the website is phished and if genuine, it prompts the user that the website is legitimate. We have detected phishing websites using Gradient Boosting Classifier with and accuracy of 97%.

Features of Proposed System

- Well-developed user interface to use regularly.
- More accurate results.
- Less Time consuming.

Advantages of Proposed System

- User Interface is Provided.
- Model is trained using many features.
- Robust in nature.

Chapter 6

SYSTEM DESIGN

System design is a transition from a user-oriented document to programmers or database personnel. The design is a solution, how to approach the creation of a new system. This is composed of several steps. It provides the understanding and procedural details necessary for implementing the system recommended in the feasibility study.

Designing goes through logical and physical stages of development, logical design reviews the present physical system, prepare input and output specification, details of the implementation plan and prepare a logical design walkthrough. Systems design is the process of defining the architecture, components, modules, interfaces, and data for a system to satisfy specified requirements. One could see it as the application of systems theory to product development.

System design is the process of defining, specifying, and creating the architecture, components, modules, interfaces, and data for a software system to satisfy specified requirements. It involves breaking down a complex system into smaller, more manageable components and defining how these components will work together to achieve the desired functionality. The goal of system design is to create a system that is reliable, scalable, maintainable, and efficient. This involves identifying the functional and non-functional requirements of the system, determining the appropriate technologies and tools to use, and creating a plan for testing and deployment.

System design typically involves creating diagrams and models to help visualize the different components of the system and how they interact with one another. Examples of such diagrams include flowcharts, data flow diagrams, and entity-relationship diagrams.

6.1 SYSTEM ARCHITECTURE

The below figure visualizes the architectural design of the project. This architecture provides the summarized view of the project. By this we can understand the rest of the project.

The Detection of Phishing Websites using Machine Learning System Architecture mainly consists of the following components:

Data Collection & Processing, Splitting Data, Feature Selection, Training Model, Gradient Boosting Classification, Model Evaluation, Prediction System, Phishing Websites Prediction.

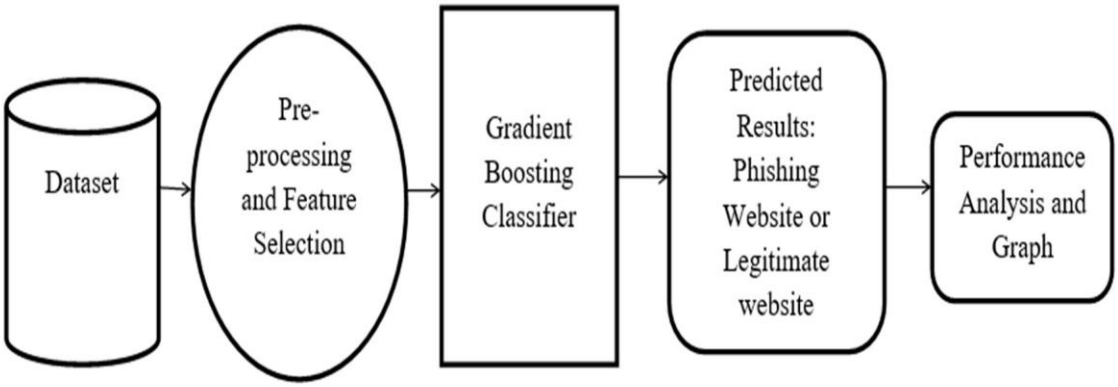


Fig 6.1: System Architecture

6.2 UML DIAGRAMS

UML (Unified Modeling Language) diagrams serve as a standardized method to visualize and document the design of a software system. They are widely used by software engineers and system architects to represent the architecture, design decisions, code structure, workflows, and interaction patterns of complex systems. UML diagrams simplify intricate programming logic into comprehensible visual models, making it easier to analyze, build, and maintain software systems.

Importance of UML Diagrams in a Disease Prediction Project

In a multi-disease prediction system, you're typically working with vast datasets, training complex machine learning models, deploying these models, and then integrating them into a user-facing application. The process involves multiple components like:

- Data collection modules
- Preprocessing pipelines
- Model training and evaluation
- User interface for input
- Prediction and output display
- Feedback or monitoring system

UML diagrams help to simplify these relationships and present them visually, enabling a clearer understanding of how each part of the system fits into the bigger picture. This is especially useful not just for developers, but also for stakeholders, testers, or collaborators who may not be familiar with the underlying code but need to understand the system workflow.

In this project six basic UML diagrams have been explained:

- Use Case Diagram
- Class Diagram

- Sequence Diagram
- Activity Diagram
- Component Diagram
- Deployment Diagram

6.2.1 Use Case Diagram

Use Case diagram is a type of behavioral diagram defined by and created from a use case analysis. A use case diagram is a type of UML (Unified Modeling Language) diagram that is used to represent the interactions between the users (or actors) and the system being designed. The use case is made up of a set of possible sequences of interactions between systems and users in a particular environment and related to a particular goal. It consists of a group of elements (for example, classes and interfaces) that can be used together in a way that will have an effect larger than the sum of the separate elements combined. The use case should contain all system activities that have significance to the users. The Use case diagram typically consists of the following components: Use cases, Actors, Relationships.

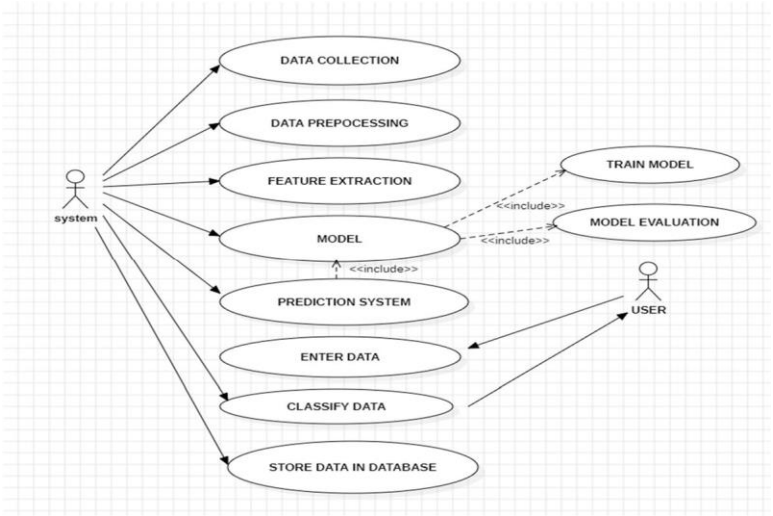


Fig. 6.2.1:Use Case Diagram

6.2.2 Class Diagram

Class diagram is a static diagram. It represents the static view of an application. Class diagram is not only used for visualizing, describing, and documenting different aspects of a system but also for constructing executable code of the software application. Class diagram describes the attributes and operations of a class and the constraints imposed on the system. The class diagrams are widely used in the modeling of object-oriented systems because they are the only UML diagrams, which can be mapped directly with object-oriented languages. Class diagram shows a collection of classes, interfaces, associations, collaborations, and constraints. It is also known as a structural diagram. Class diagrams are useful for visualizing the structure of a system and for identifying the relationships between the classes.

A Class Diagram consists of the following components: Classes, Attributes, Methods, Relationships.

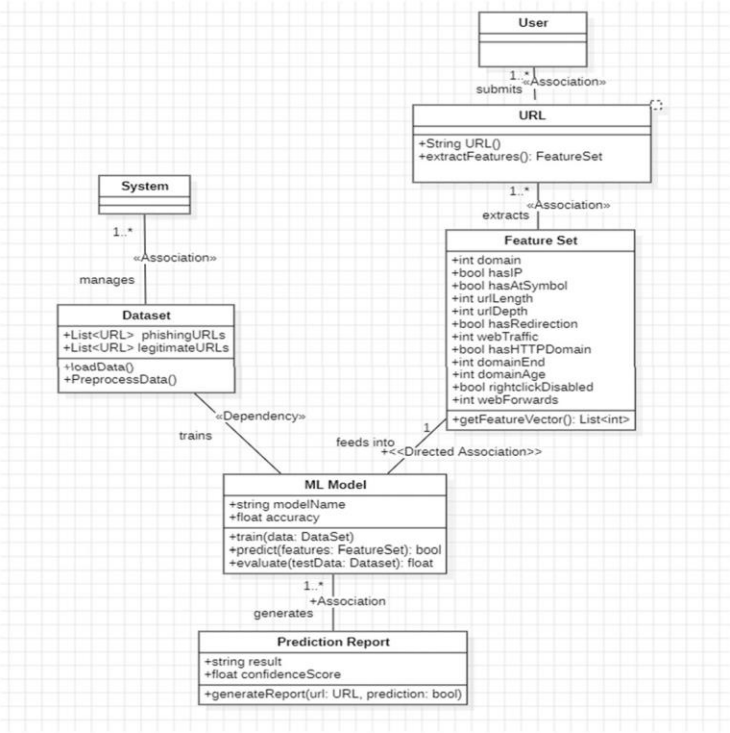


Fig. 6.2.2: Class Diagram

6.2.3 Sequence Diagram

A Sequence diagram in Unified Modelling Language (UML) is a kind of interaction diagram that shows how processes operate with one another and in what order. It is a construct of a Message Sequence Chart. A Sequence diagram depicts the sequence of actions that occur in a system. The invocation of methods in each object, and the order in which the invocation occurs is captured in a Sequence diagram. This makes the Sequence diagram a very useful tool to easily represent the dynamic behavior of a system. A sequence diagram is the most used interaction diagram. Since visualizing the interactions in a system can be a cumbersome task, we use different types of interaction diagrams to capture various features and aspects of interaction in a system.

A sequence diagram simply depicts interaction between objects in a sequential order i.e., the order in which these interactions take place. Sequence diagrams describe how and in what order the objects in a system function.

A Sequence Diagram consists of the following components: Objects, Lifelines, Messages

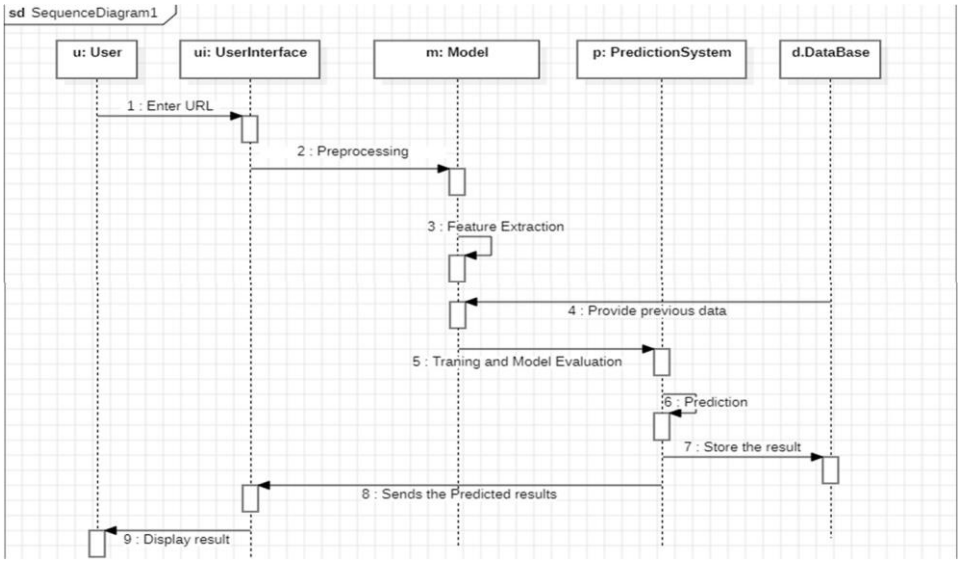


Fig. 6.2.3: Sequence Diagram

6.2.4 Activity Diagram

Activity diagram is another important diagram in UML to describe the dynamic aspects of the system. Activity diagram is basically a flowchart to represent the flow from one activity to another activity. The activity can be described as an operation of the system. An activity diagram is a type of UML (Unified Modeling Language) diagram that models the flow of activities or tasks in a system or process. It is a high-level view of the system that illustrates the steps that are required to accomplish a specific task or objective. Activity diagrams are useful for modeling the flow of activities or tasks in a system or process and for identifying potential errors or inefficiencies in the process. They can help to ensure that all the necessary activities are included in the process and that the process is designed to handle all possible scenarios. The control flow is drawn from one operation to another. This flow can be sequential, branched, or concurrent.

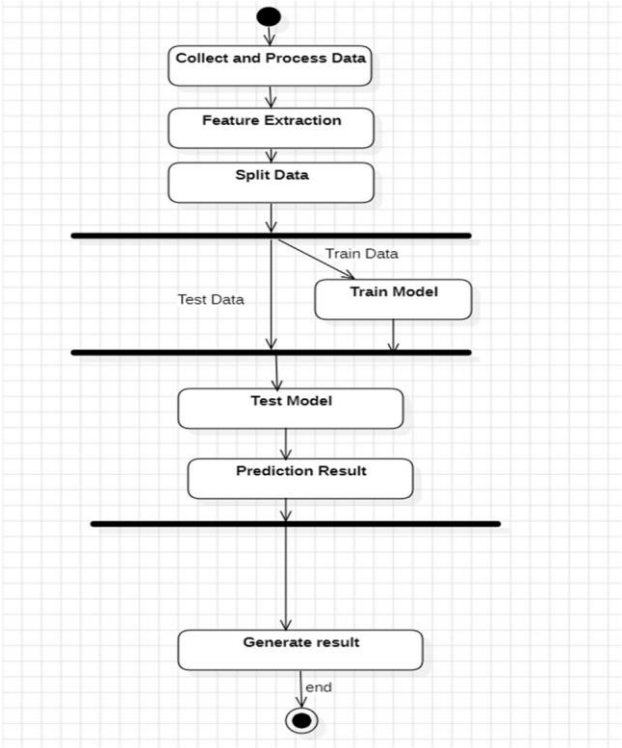


Fig. 6.2.4: Activity Diagram

6.2.5 Component Diagram

A component diagram is a type of UML (Unified Modeling Language) diagram that shows the structure of the components of a system and their interrelationships. It is a static view of the system that illustrates the physical and logical components of the system and how they work together. In a component diagram, components are represented as rectangles with the component name written inside. The interfaces of the component are shown as ports or small squares on the edges of the component. The relationships between the components are shown using connectors, such as associations or dependencies. Component diagrams are a powerful tool for system design, architecture, integration, and maintenance. They provide a clear and concise view of the system components and their relationships, which helps to ensure that the system is designed, built, and maintained correctly. Component diagrams are useful for visualizing the structure of a system and for identifying the relationships between the components.

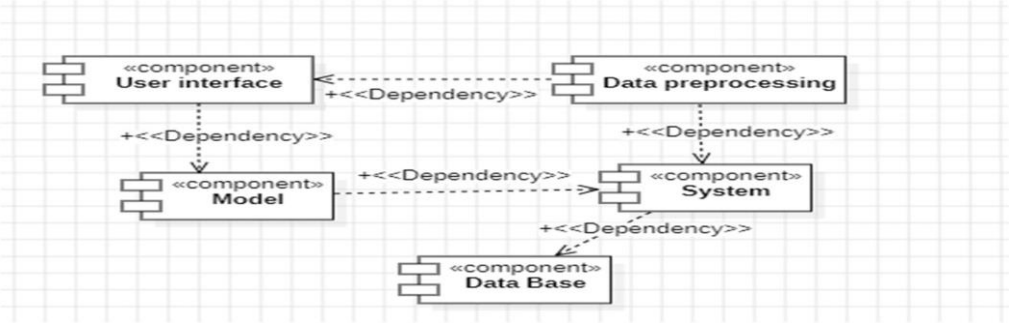


Fig. 6.2.5: Component Diagram

6.2.6 Deployment Diagram

The Deployment diagram represents the deployment view of a system. It is related to the component diagram. Because the components are deployed using the deployment diagrams. A deployment diagram consists of nodes. Nodes are nothing but physical hardware used to deploy the application. A deployment diagram is a type of UML (Unified Modeling Language) diagram that shows the physical deployment of software components

and their relationships with hardware components in a system. In a deployment diagram, components are represented as nodes or boxes, and the relationships between the components are shown using connectors, such as associations or dependencies. Nodes represent physical devices, such as servers, and components represent software modules or applications. Deployment diagrams are useful for visualizing the physical deployment of software components in a system and for identifying potential issues or constraints in the deployment.

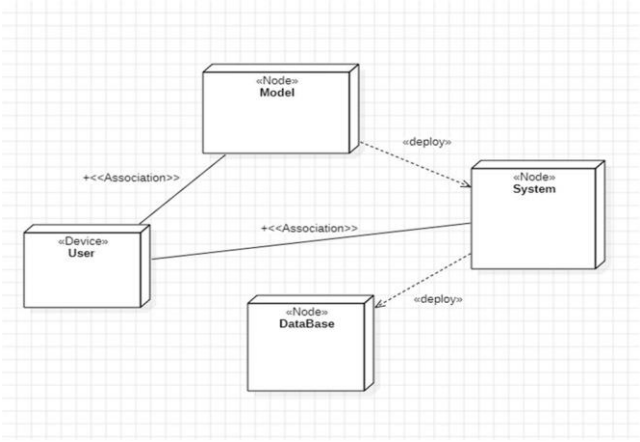


Fig. 6.2.6: Deployment Diagram

CHAPTER 7
SYSTEM IMPLEMENTATION

7.1 : ALGORITHM

Gradient Boosting Classifier Algorithm

- The Gradient Boosting Classifier is a machine learning algorithm used for binary or multi- class classification problems. It builds an ensemble of decision trees sequentially, where each tree aims to correct the errors made by the previous ones using gradient descent on a classification loss function (log-loss)
- Gradient Boosting Classifier predicts the output of a categorical dependent variable. Therefore, the outcome must be a categorical or discrete value. It can be either Yes or No, 0 or 1, true or False, etc. but instead of giving the exact value as 0 and 1, it gives the probabilistic values which lie between 0 and 1.

Understand Gradient Boosting Algorithm with example

Let's understand the intuition behind Gradient boosting with the help of an example. Here our target column is continuous hence we will use Gradient Boosting Regressor.

Step -1 The first step in gradient boosting is to build a base model to predict the observations in the training dataset. For simplicity we take an average of the target column and assume that to be the predicted value as shown below:

$$F_0(x) = \arg\min_{\gamma} \sum_{i=1}^{n} L(y_i, \gamma).$$

Why did I say we take the average of the target column? Well, there is math involved behind this. Mathematically the first step can be written as:

Looking at this may give you a headache, but don't worry we will try to understand what is written here.

Here L is our loss function Gamma is our predicted value

argmin means we have to find a predicted value/gamma for which the loss function is minimum.

Since the target column is continuous our loss function will be:

$$L = \frac{1}{n}\sum_{i=0}^{n}\left(y_i - \gamma_i\right)^2$$

Now we need to find a minimum value of gamma such that this loss function is minimum.

$$\frac{\mathrm{dL}}{\mathrm{d}\gamma} = \frac{2}{2}\left(\sum_{i=0}^{n}\left(y_i - \gamma_i\right)\right) = -\sum_{i=0}^{n}\left(y_i - \gamma_i\right)$$

We all have studied how to find minima and maxima in our 12th grade. Did we use to differentiate this loss function and then put it equal to 0 right? Yes, we will do the same here.

Let's see how to do this with the help of our example. Remember that y_i is our observed value and gamma_i is our predicted value, by plugging the values in the above formula we get:

Hence for gamma=14500, the loss function will be minimum so this value will become our prediction for the *base model*.

Step-2 The next step is to calculate the pseudo residuals which are (observed value – predicted value)

Again the question comes why only observed – predicted? Everything is mathematically proved, let's from where did this formula come from. This step can be written as:

$$r_{im} = -\left[\frac{\partial L(y_i, F(x_i))}{\partial F(x_i)}\right]_{F(x)=F_{m-1}(x)} \quad \text{for } i = 1, \ldots, n.$$

Here F(xi) is the previous model and m is the number of DT made.

We are just taking the derivative of loss function w.r.t the predicted value and we have already calculated this derivative:

$$\frac{d\mathrm{L}}{d\gamma} = -\left(y_i - \gamma_i\right) = -(Observed - Predicted)$$

$$(Observed - Predicted)$$

If you see the formula of residuals above, we see that the derivative of the loss function is multiplied by a negative sign, so now we get:

The predicted value here is the prediction made by the previous model. In our example the prediction made by the previous model (initial base model prediction) is 14500, to calculate the residuals our formula becomes:

In the next step, we will build a model on these pseudo residuals and make predictions. Why do we do this? Because we want to minimize these residuals and minimizing the residuals will eventually improve our model accuracy and prediction power. So, using the Residual as target and the original feature Cylinder number, cylinder height, and Engine location we will generate new predictions. Note that the predictions, in this case, will be the error values, not the predicted car price values since our target column is an error now. Let's say **hm(x)** is our DT made on these residuals.

Step- 4 In this step we find the output values for each leaf of our decision tree. That means there might be a case where 1 leaf gets more than 1 residual, hence we need to find the final output of all the leaves. TO find the output we can simply take the average of all the numbers in a leaf, doesn't matter if there is only 1 number or more than 1.

Let's see why do we take the average of all the numbers. Mathematically this step can be represented as:

$$\gamma_m = \arg\min_{\gamma} \sum_{i=1}^{n} L\left(y_i, F_{m-1}(x_i) + \gamma h_m(x_i)\right).$$

Here **hm(xi)** is the DT made on residuals and **m** is the number of DT.

7.2 : MODULE EXPLANATION

This project involves mainly of 6 modules. They are:

1. Data Collection and Processing
2. Splitting Data
3. Training the Model
4. Model Evaluation
5. Building the Prediction System
6. User Interface

7.2 MODULE DESCRIPTION

1. Data Collection and Processing

In the first module we develop the data collection process. This is the first real step towards the real development of a machine learning model, collecting data. This is a critical step that will cascade in how good the model will be, the more and better data that we get the better our model will perform. There are several techniques to collect the data, like web scraping, manual interventions. The dataset is referred from the popular dataset repository called kaggle.

The following is the dataset link for the Detection of Phishing Websites Using Machine Learning. Kaggle Dataset Link:

https://www.kaggle.com/datasets/jayaprakashpondy/phishing-websites-feature-dataset

2. Splitting Data

After the collection of the dataset, we split the dataset into Training Data and Testing Data. The Training Dataset is used for training the prediction model and Testing Dataset is used

for evaluating the prediction model. For this project the Data set is split as 80% and 20%. The 80% is used for training the prediction model and 20% is used for testing the prediction model.

3. Training the Model

For Training the prediction model 80% of the collected dataset is used. Logistic Regression Machine Learning Algorithm is used to train the prediction model which is more accurate.

4. Model Evaluation

For Evaluating the prediction model 20% of the collected dataset is used. The metric used for evaluating the model in the project is accuracy score. Accuracy score is the percentage of correctly predicted instances in the data set. It is calculated by dividing the number of correctly predicted instances by the total number of instances in the data set.

Algorithm	Training Accuracy	Test Accuracy
Gradient Boosting Classifier	98.9%	97.4%

Table7.2: Algorithm Accuracy

5. Building the Prediction System

The Prediction System is built using the Trained and Evaluated model. Building process of the Phishing Websites Detection System includes all the above steps such as Data Collection and Processing, splitting of the data, Training the model, Model Evaluation. The Combination of all these processes forms the Prediction System.

The Prediction System takes URL from the user and predicts whether the is Legitimate (Safe) or Phishing (UnSafe).

6. User Interface

A flexible User Interface is developed for the use of Phishing websties Prediction System. This user interface is adaptable and User friendly, JavaScript and Flask an open-source

framework for Machine learning is used to create the user interface of Prediction of phishing websites using Machine Learning. The User Interface created for Prediction of Heart Disease using Machine Learning is a webpage which takes the parameters from the user and sends to the prediction system and the result is displayed to the user.

Chapter 8

SOURCE CODE

```
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
%matplotlib inline
import seaborn as sns
from sklearn import metrics
import warnings
warnings.filterwarnings('ignore')
```

1. Loading Data:

```
data = pd.read_csv("phishing.csv")
data.head()
```

Index	UsingIP	LongURL	ShortURL	Symbol@	Redirecting//					
	PrefixSuffix-	SubDomains	HTTPS	DomainRegLen	...					
	UsingPopupWindow	IframeRedirection	AgeofDomain	DNSRecording						
	WebsiteTraffic	PageRank	GoogleIndex	LinksPointingToPage						
	StatsReport	class								
0	0	1	1	1	1	1	-1	0	1	-1	...
	1	1	-1	-1	0	-1	1	1	1	-1	
1	1	1	0	1	1	1	-1	-1	-1	-1	...
	1	1	1	-1	1	-1	1	0	-1	-1	
2	2	1	0	1	1	1	-1	-1	-1	1	...
	1	1	-1	-1	1	-1	1	-1	1	-1	
3	3	1	0	-1	1	1	-1	1	1	-1	...
	-1	1	-1	-1	0	-1	1	1	1	1	

4	4	-1	0	-1	1	-1	-1	1	1	-1	...
	1	1	1	1	1	-1	1	-1	-1	1	

5 rows × 32 columns

2. Familiarizing with Data & EDA:

In this step, few dataframe methods are used to look into the data and its features.

```
data.shape
```

(11054, 32)

```
data.columns
```

```
Index(['Index', 'UsingIP', 'LongURL', 'ShortURL', 'Symbol@', 'Redirecting//',
       'PrefixSuffix-', 'SubDomains', 'HTTPS', 'DomainRegLen', 'Favicon',
       'NonStdPort', 'HTTPSDomainURL', 'RequestURL', 'AnchorURL',
       'LinksInScriptTags', 'ServerFormHandler', 'InfoEmail', 'AbnormalURL',
       'WebsiteForwarding', 'StatusBarCust', 'DisableRightClick',
       'UsingPopupWindow', 'IframeRedirection', 'AgeofDomain', 'DNSRecording',
       'WebsiteTraffic', 'PageRank', 'GoogleIndex', 'LinksPointingToPage',
       'StatsReport', 'class']
```

```
data.info()
```

```
<class  'pandas.core.frame.DataFrame'>
RangeIndex: 11054 entries, 0 to 11053
Data columns (total 32 columns):
 #   Column              Non-Null Count  Dtype
---  -------             ----------------  ------
 0   Index               11054 non-null  int64
 1   UsingIP             11054 non-null  int64
 2   LongURL             11054 non-null  int64
 3   ShortURL            11054 non-null  int64
 4   Symbol@             11054 non-null  int64
 5   Redirecting//       11054 non-null  int64
```

```
 6   PrefixSuffix-         11054 non-null  int64
 7   SubDomains            11054 non-null  int64
 8   HTTPS                 11054 non-null  int64
 9   DomainRegLen          11054 non-null  int64
 10  Favicon               11054 non-null  int64
 11  NonStdPort            11054 non-null  int64
 12  HTTPSDomainURL        11054 non-null  int64
 13  RequestURL            11054 non-null  int64
 14  AnchorURL             11054 non-null  int64
 15  LinksInScriptTags     11054 non-null  int64
 16  ServerFormHandler     11054 non-null  int64
 17  InfoEmail             11054 non-null  int64
 18  AbnormalURL           11054 non-null  int64
19  WebsiteForwarding     11054 non-null  int64
 20  StatusBarCust         11054 non-null  int64
 21  DisableRightClick     11054 non-null  int64
 22  UsingPopupWindow      11054 non-null  int64
 23  IframeRedirection     11054 non-null  int64
 24  AgeofDomain           11054 non-null  int64
 25  DNSRecording          11054 non-null  int64
 26  WebsiteTraffic        11054 non-null  int64
 27  PageRank              11054 non-null  int64
 28  GoogleIndex           11054 non-null  int64
 29  LinksPointingToPage  11054 non-null  int64
 30  StatsReport           11054 non-null  int64
 31  class                 11054 non-null  int64
dtypes: int64(32)
memory usage: 2.7 MB
```

```
# nunique value in columns
data.nunique()
Index                11054
UsingIP                2
LongURL                3
ShortURL               2
Symbol@                2
Redirecting//          2
PrefixSuffix-          2
SubDomains             3
HTTPS                  3
DomainRegLen           2
Favicon                2
NonStdPort             2
HTTPSDomainURL         2
RequestURL             2
AnchorURL              3
LinksInScriptTags      3
ServerFormHandler      3
InfoEmail              2
AbnormalURL            2
WebsiteForwarding      2
StatusBarCust          2
DisableRightClick      2
UsingPopupWindow       2
IframeRedirection      2
AgeofDomain            2
DNSRecording           2
```

WebsiteTraffic 3
PageRank 2
GoogleIndex 2
LinksPointingToPage 3
StatsReport 2
class 2
dtype: int64
data.describe().T

	count	mean	std	min	25%	50%	75%	max
UsingIP	11054.0	0.313914	0.949495	-1.0	-1.0	1.0	1.0	1.0
LongURL	11054.0	-0.633345	0.7integration973	-1.0	-1.0	-1.0	-1.0	1.0
ShortURL	11054.0	0.738737	0.674024	-1.0	1.0	1.0	1.0	1.0
Symbol@	11054.0	0.700561	0.713625	-1.0	1.0	1.0	1.0	1.0
Redirecting//	11054.0	0.741632	0.670837	-1.0	1.0	1.0	1.0	1.0
PrefixSuffix-	11054.0	-0.734938	0.678165	-1.0	-1.0	-1.0	-1.0	1.0
SubDomains	11054.0	0.064049	0.817492	-1.0	-1.0	0.0	1.0	1.0
HTTPS	11054.0	0.251040	0.911856	-1.0	-1.0	1.0	1.0	1.0
DomainRegLen	11054.0	-0.336711	0.941651	-1.0	-1.0	-1.0	1.0	1.0

Favicon	11054.0	0.628551	0.777804	-1.0	1.0	1.0	1.0
1.0							
NonStdPort	11054.0	0.728243	0.685350	-1.0	1.0	1.0	1.0
1.0							
HTTPSDomainURL	11054.0	0.675231	0.737640	-1.0	1.0	1.0	
1.0	1.0						
RequestURL	11054.0	0.186720	0.982458	-1.0	-1.0	1.0	1.0
1.0							
AnchorURL	11054.0	-0.076443	0.715116	-1.0	-1.0	0.0	0.0
1.0							
LinksInScriptTags	11054.0	-0.118238	0.763933	-1.0	-1.0	0.0	
0.0	1.0						
ServerFormHandler	11054.0	-0.595712	0.759168	-1.0	-1.0	-1.0	
-1.0	1.0						
InfoEmail	11054.0	0.635788	0.771899	-1.0	1.0	1.0	1.0
1.0							
AbnormalURL	11054.0	0.705446	0.708796	-1.0	1.0	1.0	
1.0	1.0						
WebsiteForwarding	11054.0	0.115705	0.319885	0.0	0.0	0.0	
0.0	1.0						
StatusBarCust	11054.0	0.762077	0.647516	-1.0	1.0	1.0	1.0
1.0							
DisableRightClick	11054.0	0.913877	0.406009	-1.0	1.0	1.0	
1.0	1.0						
UsingPopupWindow	11054.0	0.613353	0.789845	-1.0	1.0	1.0	
1.0	1.0						
IframeRedirection	11054.0	0.816899	0.576807	-1.0	1.0	1.0	
1.0	1.0						

AgeofDomain	11054.0	0.061335	0.998162	-1.0	-1.0	1.0	1.0
1.0							
DNSRecording	11054.0	0.377239	0.926158	-1.0	-1.0	1.0	
1.0	1.0						
WebsiteTraffic	11054.0	0.287407	0.827680	-1.0	0.0	1.0	
1.0	1.0						
PageRank	11054.0	-0.483626	0.875314	-1.0	-1.0	-1.0	1.0
1.0							
GoogleIndex	11054.0	0.721549	0.692395	-1.0	1.0	1.0	1.0
1.0							
LinksPointingToPage	11054.0	0.343948	0.569936	-1.0	0.0	0.0	
1.0	1.0						
StatsReport	11054.0	0.719739	0.694276	-1.0	1.0	1.0	1.0
1.0							
class	11054.0	0.113986	0.993527	-1.0	-1.0	1.0	1.0	1.0

data_set.append(9 OBSERVATIONS:

There are 11054 instances and 31 fearures in dataset.

Out of which 30 are independent features where as 1 is dependent feature.

Each feature is in int datatype, so there is no need to use LabelEncoder.

There is no outlier present in dataset.

There is no missing value in dataset.

3. Visualizing the data:

Few plots and graphs are displayed to find how the data is distributed and the how features are related to each other.

```
plt.figure(figsize=(15,15))
sns.heatmap(data.corr(), annot=True)
plt.show()
```

```
df = data[['PrefixSuffix-', 'SubDomains', 'HTTPS','AnchorURL','WebsiteTraffic','class']]
sns.pairplot(data = df,hue="class",corner=True);
data['class'].value_counts().plot(kind='pie',autopct='%1.2f%%')
plt.title("Phishing Count")
plt.show()
```

The data is split into train & test sets, 80-20 split.

```
X = data.drop(["class"],axis =1)
y = data["class"]
# Spliting the dataset into train and test sets: 80-20 split
from sklearn.model_selection import train_test_split
X_train, X_test, y_train, y_test = train_test_split(X, y, test_size = 0.2, random_state = 42)
X_train.shape, y_train.shape, X_test.shape, y_test.shape
((8843, 30), (8843,), (2211, 30), (2211,))
```

5. Model Building & Training:

Gradient Boosting

The metrics considered to evaluate the model performance are Accuracy & F1 score.

```
# Creating holders to store the model performance results
ML_Model = []
accuracy = []
f1_score = []
recall = []
precision = []
#function to call for storing the results
def storeResults(model, a,b,c,d):
ML_Model.append(model)
accuracy.append(round(a, 3))
  f1_score.append(round(b, 3))
  recall.append(round(c, 3))
```

```
precision.append(round(d,  3))
```

5.7.Gradient Boosting Classifier

Gradient boosting classifiers are a group of machine learning algorithms that combine many weak learning models together to create a strong predictive model. Decision trees are usually used when doing gradient boosting. Boosting algorithms play a crucial role in dealing with bias variance trade-off. Unlike bagging algorithms, which only controls for high variance in a model, boosting controls both the aspects (bias & variance), and is considered to be more effective.

```
from sklearn.ensemble import GradientBoostingClassifier
gbc = GradientBoostingClassifier(max_depth=4,learning_rate=0.7)
# fit the model
gbc.fit(X_train,y_train)
GradientBoostingClassifier
GradientBoostingClassifier(learning_rate=0.7, max_depth=4)
#predicting the target value from the model for the samples
y_train_gbc = gbc.predict(X_train)
y_test_gbc = gbc.predict(X_test)
#computing the accuracy, f1_score, Recall, precision of the model performance
acc_train_gbc = metrics.accuracy_score(y_train,y_train_gbc)
acc_test_gbc = metrics.accuracy_score(y_test,y_test_gbc)
print("Gradient Boosting Classifier : Accuracy on training Data: {:.3f}".format(acc_train_gbc))
print("Gradient Boosting Classifier : Accuracy on test Data: {:.3f}".format(acc_test_gbc))
print()
f1_score_train_gbc = metrics.f1_score(y_train,y_train_gbc)
f1_score_test_gbc = metrics.f1_score(y_test,y_test_gbc)
print("Gradient Boosting Classifier : f1_score on training Data: {:.3f}".format(f1_score_train_gbc))
```

```
print("Gradient Boosting Classifier : f1_score on test Data: {:.3f}".format(f1_score_test_gbc))
print()
recall_score_train_gbc = metrics.recall_score(y_train,y_train_gbc)
recall_score_test_gbc =  metrics.recall_score(y_test,y_test_gbc)
print("Gradient Boosting Classifier : Recall on training Data: {:.3f}".format(recall_score_train_gbc))
print("Gradient Boosting Classifier : Recall on test Data: {:.3f}".format(recall_score_test_gbc))
print()
precision_score_train_gbc = metrics.precision_score(y_train,y_train_gbc)
precision_score_test_gbc = metrics.precision_score(y_test,y_test_gbc)
print("Gradient Boosting Classifier : precision on training Data: {:.3f}".format(precision_score_train_gbc))
print("Gradient Boosting Classifier : precision on test Data: {:.3f}".format(precision_score_test_gbc))
```

```
Gradient Boosting Classifier : Accuracy on training Data: 0.989
Gradient Boosting Classifier : Accuracy on test Data: 0.974
Gradient Boosting Classifier : f1_score on training Data: 0.990
Gradient Boosting Classifier : f1_score on test Data: 0.977
Gradient Boosting Classifier : Recall on training Data: 0.994
Gradient Boosting Classifier : Recall on test Data: 0.989
Gradient Boosting Classifier : precision on training Data: 0.986
Gradient Boosting Classifier : precision on test Data: 0.966
```

```
#computing the classification report of the model
print(metrics.classification_report(y_test, y_test_gbc))
```

```
              precision    recall  f1-score   support

          -1       0.99      0.96      0.97       976
```

```
           1       0.97      0.99      0.98      1235

    accuracy                           0.97      2211
   macro avg       0.98      0.97      0.97      2211
weighted avg       0.97      0.97      0.97      2211
from sklearn.metrics import confusion_matrix
print(confusion_matrix(y_test, y_test_gbc))
[[ 933   43]
 [  14 1221]]
from sklearn.metrics import confusion_matrix
cm=confusion_matrix(y_test, y_test_gbc)
cm
array([[ 933,   43],
       [  14, 1221]], dtype=int64)
import seaborn as sns
import matplotlib.pyplot as plt
f, ax=plt.subplots(figsize=(5,5))
sns.heatmap(cm,annot=True,linewidths=0.5,linecolor="red",fmt=".0f",ax=ax)
plt.xlabel("y_pred")
plt.ylabel("y_true")
plt.show()
training_accuracy = []
test_accuracy = []
# try learning_rate from 0.1 to 0.9
depth = range(1,10)
for n in depth:
    forest_test =  GradientBoostingClassifier(learning_rate = n*0.1)
    forest_test.fit(X_train, y_train)
    # record training set accuracy
```

```
    training_accuracy.append(forest_test.score(X_train, y_train))
    test_accuracy.append(forest_test.score(X_test, y_test))
plt.figure(figsize=None)
plt.plot(depth, training_accuracy, label="training accuracy")
plt.plot(depth, test_accuracy, label="test accuracy")
plt.ylabel("Accuracy")
plt.xlabel("learning_rate")
plt.legend();
training_accuracy = []
test_accuracy = []
# try learning_rate from 0.1 to 0.9
depth = range(1,10,1)
for n in depth:
    forest_test =  GradientBoostingClassifier(max_depth=n,learning_rate = 0.7)
    forest_test.fit(X_train, y_train)
    # record training set accuracy
    training_accuracy.append(forest_test.score(X_train, y_train))
    # record generalization accuracy
    test_accuracy.append(forest_test.score(X_test, y_test))
plt.figure(figsize=None)
plt.plot(depth, training_accuracy, label="training accuracy")
plt.plot(depth, test_accuracy, label="test accuracy")
plt.ylabel("Accuracy")
plt.xlabel("max_depth")
plt.legend();
storeResults('Gradient Boosting Classifier',acc_test_gbc,f1_score_test_gbc,
            recall_score_train_gbc,precision_score_train_gbc)
# instantiate the model
```

```
gbc = GradientBoostingClassifier(max_depth=4,learning_rate=0.7)
# fit the model
gbc.fit(X_train,y_train)
GradientBoostingClassifier
GradientBoostingClassifier(learning_rate=0.7, max_depth=4)
import pickle
pickle.dump(gbc, open('model.pkl', 'wb'))
#checking the feature improtance in the model
plt.figure(figsize=(9,7))
n_features = X_train.shape[1]
plt.barh(range(n_features), gbc.feature_importances_, align='center')
plt.yticks(np.arange(n_features), X_train.columns)
plt.title("Feature importances using permutation on full model")
plt.xlabel("Feature importance")
plt.ylabel("Feature")
plt.show()
```

FRONTEND CODE

JAVASCRIPT CODE:

```
(function ($) {
    "use strict";
    // Spinner
    var spinner = function () {
        setTimeout(function () {
            if ($('#spinner').length > 0) {
                $('#spinner').removeClass('show');
            }
        }, 1);
    };
```

```
spinner()
// Initiate the wowjs
new WOW().init();
// Sticky Navbar
$(window).scroll(function () {
    if ($(this).scrollTop() > 45) {
        $('.navbar').addClass('sticky-top shadow-sm');
    } else {
        $('.navbar').removeClass('sticky-top shadow-sm');
    }
});
// Dropdown on mouse hover
const $dropdown = $(".dropdown");
const $dropdownToggle = $(".dropdown-toggle");
const $dropdownMenu = $(".dropdown-menu");
const showClass = "show";
$(window).on("load resize", function() {
    if (this.matchMedia("(min-width: 992px)").matches) {
        $dropdown.hover(
        function() {
            const $this = $(this);
            $this.addClass(showClass);
            $this.find($dropdownToggle).attr("aria-expanded", "true");
            $this.find($dropdownMenu).addClass(showClass);
        },
        function() {
            const $this = $(this);
            $this.removeClass(showClass);
```

```
            $this.find($dropdownToggle).attr("aria-expanded",  "false");
            $this.find($dropdownMenu).removeClass(showClass);
        }
        );
    } else {
        $dropdown.off("mouseenter mouseleave");
    }
});
$('[data-toggle="counter-up"]').counterUp({
    delay: 10,
    time: 2000
});
$(window).scroll(function () {
    if ($(this).scrollTop() > 100) {
        $('.back-to-top').fadeIn('slow');
    } else {
        $('.back-to-top').fadeOut('slow');
    }
});
$('.back-to-top').click(function () {
    $('html, body').animate({scrollTop: 0}, 1500, 'easeInOutExpo');
    return false;
});
({
    autoplay: true,
    smartSpeed: 1500,
    dots: true,
    loop: true,
```

```
        center: true,
        responsive: {
            0:{
                items:1
            },
            576:{
                items:1
            },
            768:{
                items:2
            },
            992:{
                items:3
            }
        }
    });
    $('.vendor-carousel').owlCarousel({
        loop: true,
        margin: 45,
        dots: false,
        loop: true,
        autoplay: true,
        smartSpeed: 1000,
        responsive: {
            0:{
                items:2
            },
            576:{
```

```
            items:4
        },
        768:{
            items:6
        },
        992:{
            items:8
        }
    }
  });
})(jQuery);
```

Page 4:Prediction

```
<html lang="en">
<head>
<meta charset="utf-8">
<title>URL</title>
<meta content="width=device-width, initial-scale=1.0" name="viewport">
<meta content="Free HTML Templates" name="keywords">
<meta content="Free HTML Templates" name="description">
<!-- Favicon -->
<link href="../static/img/favicon.ico" rel="icon">
<!-- Google Web Fonts -->
<link rel="preconnect" href="https://fonts.googleapis.com">
<link rel="preconnect" href="https://fonts.gstatic.com" crossorigin>
<link
href="https://fonts.googleapis.com/css2?family=Nunito:wght@400;600;700;800&family
=Rubik:wght @400;500;600;700&display=swap" rel="stylesheet">
<!-- Icon Font Stylesheet -->
```

```
<link href="https://cdnjs.cloudflare.com/ajax/libs/font-awesome/5.10.0/css/all.min.css" rel="stylesheet">

<link href="https://cdn.jsdelivr.net/npm/bootstrap-icons@1.4.1/font/bootstrap-icons.css" rel="stylesheet">

<!-- Libraries Stylesheet -->

<link href="../static/lib/owlcarousel/assets/owl.carousel.min.css" rel="stylesheet">

<link href="../static/lib/animate/animate.min.css" rel="stylesheet">

<!-- Customized Bootstrap Stylesheet -->

<link href="../static/css/bootstrap.min.css" rel="stylesheet">

<!-- Template Stylesheet -->

<link href="../static/css/result.css" rel="stylesheet">

</head>

<body>

<!-- Spinner Start -->

<div id="spinner" class="show bg-white position-fixed translate-middle w-100 vh-100 top-50 start- 50 d-flex align-items-center justify-content-center">

<div class="spinner"></div>

</div>

<!-- Spinner End -->

<!-- Topbar Start -->

<!-- Topbar End -->

<!-- Navbar Start -->

<div class="container-fluid position-relative p-0">

<nav class="navbar navbar-expand-lg navbar-dark px-5 py-3 py-lg-0">

<a href="{{url_for('first')}}" class="navbar-brand p-0">

<h1 class="m-0">Phishing Websites</h1>

</a>

<button class="navbar-toggler" type="button" data-bs-toggle="collapse" data-bs-target="#navbarCollapse">

<span class="fa fa-bars"></span>
```

```
</button>
<div class="collapse navbar-collapse" id="navbarCollapse">
<div class="navbar-nav ms-auto py-0">
<a href="{{url_for('first')}}" class="nav-item nav-link">Home</a>
<a href="{{url_for('login')}}" class="nav-item nav-link">Login</a>
<a href="{{url_for('upload')}}" class="nav-item nav-link">Upload</a>
<a href="{{url_for('index')}}" class="nav-item nav-link active">Prediction</a>
<a href="{{url_for('performance')}}" class="nav-item nav-
link">Performance_Analysis</a>
</div>
</div>
</nav>
<div class="container-fluid bg-primary py-5 bg-header" style="margin-bottom: 90px;">
<div class="row py-5">
<div class="col-12 pt-lg-5 mt-lg-5 text-center">
<h1 class="display-4 text-white animated zoomIn">Prediction</h1>
<i class="far fa-circle text-white px-2"></i>
<a href="" class="h5 text-white">Phishing Websites</a>
</div>
</div>
</div>
<!-- Navbar End -->
<!-- Full Screen Search Start -->
<div class="modal fade" id="searchModal" tabindex="-1">
<div class="modal-dialog modal-fullscreen">
<div class="modal-content" style="background: rgba(9, 30, 62, .7);">
<div class="modal-header border-0">
<button type="button" class="btn bg-white btn-close" data-bs-dismiss="modal" aria-
label="Close"></button>
```

```
</div>
<div class="modal-body d-flex align-items-center justify-content-center">
<div class="input-group" style="max-width: 600px;">
<input type="text" class="form-control bg-transparent border-primary p-3" placeholder="Type search keyword">
<button class="btn btn-primary px-4"><i class="bi bi-search"></i></button>
</div>
</div>
</div>
</div>
</div>
<!-- Full Screen Search End -->
<!-- Team Start -->
<div class="container-fluid py-5 wow fadeInUp" data-wow-delay="0.1s">
<div class="container py-5">
<div class="section-title text-center position-relative pb-3 mb-5 mx-auto" style="max-width: 600px;">
<h5 class="fw-bold text-primary text-uppercase">PHISHING URL DETECTION</h5>
<class="mb-0">Result</h1>
</div>
<body>
<div class=" container">
</div>
<center>
<div class="col-md" id="form2">
<br>
<h6 class = "right "><a href= {{ url }} target="_blank">{{ url }}</a></h6>
<br>
<h3 id="prediction"></h3>
```

```
</div> </center>
</div>
<br>
</div>
<!-- JavaScript -->
<script src="https://code.jquery.com/jquery-3.5.1.slim.min.js" integrity="sha384-
DfXdz2htPH0lsSSs5nCTpuj/zy4C+OGpamoFVy38MVBnE+IbbVYUew+OrCXaRkfj"
crossorigin="anonymous"></script>
<script src="https://cdn.jsdelivr.net/npm/popper.js@1.16.0/dist/umd/popper.min.js"
integrity="sha384-
Q6E9RHvbIyZFJoft+2mJbHaEWldlvI9IOYy5n3zV9zzTtmI3UksdQRVvoxMfooAo"
crossorigin="anonymous"></script>
<script src="https://stackpath.bootstrapcdn.com/bootstrap/4.5.0/js/bootstrap.min.js"
integrity="sha384-
OgVRvuATP1z7JjHLkuOU7Xw704+h835Lr+6QL9UvYjZE3Ipu6Tp75j7Bh/kR0JKI"
crossorigin="anonymous"></script>
<script>
let x = '{{xx}}'; let num = x*100;
if (0<=x && x<0.50)
num = 100-num;
}
let txtx = num.toString(); if(x<=1 && x>=0.50){
var label = "This Website is safe to use...";
document.getElementById("prediction").innerHTML = label;
document.getElementById("button1").style.display="block";
}
else if (0<=x && x<0.50){
var label = "This Website is may be unsafe to use..."
document.getElementById("prediction").innerHTML = label ;
document.getElementById("button2").style.display="block";
}
```

```
script>
<!-- JavaScript -->
</body>
</div>
</div>
<!-- Team End -->
<!-- Vendor Start -->
<!-- Footer Start -->
<div class="container-fluid bg-dark text-light mt-5 wow fadeInUp" data-wow-delay="0.1s">
<div class="container">
</div>
</div>
<div class="container-fluid text-white" style="background: #061429;">
<div class="container text-center">
<div class="row justify-content-end">
<div class="col-lg-8 col-md-6">
<div class="d-flex align-items-center justify-content-center" style="height: 75px;">
<p class="mb-0"> <a class="text-white border-bottom" href="#">
<!--/*** This template is free as long as you keep the footer author’s credit link/attribution link/backlink. If you'd like to use the template without the footer author’s credit link/attribution link/backlink, you can purchase the Credit Removal License from "https://htmlcodex.com/credit-removal". Thank you for your support. ***/-->
<a class="text-white border-bottom" href="https://htmlcodex.com"></a></p>
</div>
</div>
</div>
</div>
</div>
<!-- Footer End -->
```

<!-- Back to Top -->

<!-- JavaScript Libraries -->

<script src="https://code.jquery.com/jquery-3.4.1.min.js"></script>

<script src="https://cdn.jsdelivr.net/npm/bootstrap@5.0.0/dist/js/bootstrap.bundle.min.js"></script>

<script src="../static/lib/wow/wow.min.js"></script>

<script src="../static/lib/easing/easing.min.js"></script>

<script src="../static/lib/waypoints/waypoints.min.js"></script>

<script src="../static/lib/counterup/counterup.min.js"></script>

<script src="../static/lib/owlcarousel/owl.carousel.min.js"></script>

<!-- Template Javascript -->

<script src="../static/js/main.js"></script>

</body>

</html>

Page 5: Activites

:root {

--primary: #06A3DA;

--secondary: #34AD54;

--light: #EEF9FF;

--dark: #091E3E;

}

/*** Spinner ***/

.spinner { width: 40px; height: 40px;

background: var(--primary); margin: 100px auto;

-webkit-animation: sk-rotateplane 1.2s infinite ease-in-out; animation: sk-rotateplane 1.2s infinite ease-in-out;

}@-webkit-keyframes sk-rotateplane { 0% {

```
-webkit-transform: perspective(120px)
}
50% {
-webkit-transform: perspective(120px) rotateY(180deg)
}
100% {
-webkit-transform: perspective(120px) rotateY(180deg) rotateX(180deg)
}
}
@keyframes sk-rotateplane { 0% {
transform: perspective(120px) rotateX(0deg) rotateY(0deg);
-webkit-transform: perspective(120px) rotateX(0deg) rotateY(0deg)
}
50% {
transform: perspective(120px) rotateX(-180.1deg) rotateY(0deg);
-webkit-transform: perspective(120px) rotateX(-180.1deg) rotateY(0deg)
}
100% {
transform: perspective(120px) rotateX(-180deg) rotateY(-179.9deg);
-webkit-transform: perspective(120px) rotateX(-180deg) rotateY(-179.9deg);
}}
#spinner { opacity: 0;
visibility: hidden;
transition: opacity .5s ease-out, visibility 0s linear .5s; z-index: 99999;
}
#spinner.show {
transition: opacity .5s ease-out, visibility 0s linear 0s; visibility: visible;
opacity: 1;
```

```
}
/*** Heading ***/ h1,
h2,
.fw-bold {
font-weight: 800 !important;
}
h3,
h4,
.fw-semi-bold {
font-weight: 700 !important;
}
h5,
h6,
.fw-medium {
font-weight: 600 !important;
}



/*** Button ***/
.btn {
font-family: 'Nunito', sans-serif; font-weight: 600;
transition: .5s;
}
.btn-primary,
.btn-secondary { color: #FFFFFF;
box-shadow: inset 0 0 0 50px transparent;
}
```

```
.btn-primary:hover {
box-shadow: inset 0 0 0 0 var(--primary);
}
.btn-secondary:hover {
box-shadow: inset 0 0 0 0 var(--secondary);
}
.btn-square { width: 36px; height: 36px;
.btn-sm-square { width: 30px; height: 30px;
}
.btn-lg-square { width: 48px; height: 48px;}
.btn-square,
.btn-sm-square,
.btn-lg-square { padding-left: 0;
padding-right: 0;
text-align: center;
}
/*** Navbar ***/
.navbar-dark .navbar-nav .nav-link { font-family: 'Nunito', sans-serif; position: relative;
margin-left: 25px; padding: 35px 0; color: #FFFFFF; font-size: 18px; font-weight: 600;
outline: none; transition: .5s;
}
.sticky-top.navbar-dark .navbar-nav .nav-link { padding: 20px 0;
color: var(--dark);
}
.navbar-dark .navbar-nav .nav-link:hover,
.navbar-dark .navbar-nav .nav-link.active { color: var(--primary);
}
.navbar-dark .navbar-brand h1 { color: #FFFFFF;
}
```

```
.navbar-dark .navbar-toggler { color: var(--primary) !important;
border-color: var(--primary) !important;
}
@media (max-width: 991.98px) {
.sticky-top.navbar-dark { position: relative;
: #FFFFFF;
}
.navbar-dark .navbar-nav .nav-link,
.navbar-dark .navbar-nav .nav-link.show,
.sticky-top.navbar-dark .navbar-nav .nav-link { padding: 10px 0;
color: var(--dark);
}
.navbar-dark .navbar-brand h1 { color: var(--primary);
}
}
@media (min-width: 992px) {
.navbar-dark { position: absolute; width: 100%;
top: 0;
left: 0;
border-bottom: 1px solid rgba(256, 256, 256, .1);
z-index: 999;
}
.sticky-top.navbar-dark { position: fixed; background: #FFFFFF;
}
.navbar-dark .navbar-nav .nav-link::before { position: absolute;
content: ""; width: 0; height: 2px; bottom: -1px; left: 50%;
background: var(--primary); transition: .5s;
}
```

```
.navbar-dark .navbar-nav .nav-link:hover::before,
.navbar-dark .navbar-nav .nav-link.active::before { width: 100%;
left: 0;
}
.navbar-dark .navbar-nav .nav-link.nav-contact::before { display: none;
.sticky-top.navbar-dark .navbar-brand h1 { color: var(--primary);
}
}
/*** Carousel ***/
.carousel-caption { top: 0;
left: 0;
right: 0;
bottom: 0;
background: rgba(9, 30, 62, .7);
z-index: 1;
}
@media (max-width: 576px) {
.carousel-caption h5 { font-size: 14px;
font-weight: 500 !important;
}
.carousel-caption h1 { font-size: 30px;
font-weight: 600 !important;
}




}
.carousel-control-prev,
```

```
.carousel-control-next { width: 10%;
}
.carousel-control-prev-icon,
.carousel-control-next-icon { width: 3rem;
height: 3rem;
}
/*** Section Title ***/
.section-title::before { position: absolute; content: "";
width: 150px; height: 5px; left: 0;
bottom: 0;
background: var(--primary); border-radius: 2px;
}
.section-title.text-center::before { left: 50%;
margin-left: -75px;
}
.section-title.section-title-sm::before { width: 90px;
height: 3px;
}
.section-title::after { position: absolute; content: "";
width: 6px; height: 5px; bottom: 0px;
background: #FFFFFF;
-webkit-animation: section-title-run 5s infinite linear; animation: section-title-run 5s
infinite linear;
}
.section-title.section-title-sm::after { width: 4px;
height: 3px;
}
.section-title.text-center::after {
```

```
-webkit-animation: section-title-run-center 5s infinite linear; animation: section-title-run-center 5s infinite linear;

}

.section-title.section-title-sm::after {

-webkit-animation: section-title-run-sm 5s infinite linear; animation: section-title-run-sm 5s infinite linear;

}

@-webkit-keyframes section-title-run {

0% {left: 0; } 50% { left : 145px; } 100% { left: 0; }

}

@-webkit-keyframes section-title-run-center {

0% { left: 50%; margin-left: -75px; } 50% { left : 50%; margin-left: 45px; } 100% { left: 50%; margin- left: -75px; }

}

@-webkit-keyframes section-title-run-sm {

0% {left: 0; } 50% { left : 85px; } 100% { left: 0; }

}

/*** Service ***/

.service-item { position: relative;

height: 300px; padding: 0 30px; transition: .5s;

}

.service-item .service-icon { margin-bottom: 30px; width: 60px;

height: 60px; display: flex;

align-items: center; justify-content: center;

background: var(--primary); border-radius: 2px; transform: rotate(-45deg);

}

.service-item .service-icon i { transform: rotate(45deg);

}

.service-item a.btn { position: absolute; width: 60px; bottom: -48px; left: 50%;
```

```
margin-left: -30px; opacity: 0;
}
.service-item:hover a.btn { bottom: -24px;
opacity: 1;
}
/*** Testimonial ***/
.testimonial-carousel .owl-dots {
margin-top: 15px; display: flex;
align-items: flex-end; justify-content: center;
}
.testimonial-carousel .owl-dot { position: relative;
display: inline-block; margin: 0 5px; width: 15px;
height: 15px; background: #DDDDDD; border-radius: 2px; transition: .5s;
}
.testimonial-carousel .owl-dot.active { width: 30px;
background: var(--primary);
}
.testimonial-carousel .owl-item.center { position: relative;
z-index: 1;
}
.testimonial-carousel .owl-item .testimonial-item { transition: .5s;
}
.testimonial-carousel .owl-item.center .testimonial-item { background: #FFFFFF
!important;
box-shadow: 0 0 30px #DDDDDD;
}
/*** Team ***/
.team-item {
```

```
transition: .5s;
}

.team-social { position: absolute; width: 100%;
height: 100%;
top: 0;
left: 0; display: flex;
align-items: center; justify-content: center; transition: .5s;
}
.team-social a.btn { position: relative; margin: 0 3px; margin-top: 100px; opacity: 0;
}
.team-item:hover {
box-shadow: 0 0 30px #DDDDDD;
}
.team-item:hover .team-social { background: rgba(9, 30, 62, .7);
}
.team-item:hover .team-social a.btn:first-child { opacity: 1;
margin-top: 0; transition: .3s 0s;
}
.team-item:hover .team-social a.btn:nth-child(2) {
opacity: 1;
margin-top: 0; transition: .3s .05s;
}
.team-item:hover .team-social a.btn:nth-child(3) { opacity: 1;
margin-top: 0; transition: .3s .1s;
```

```
}

.team-item:hover .team-social a.btn:nth-child(4) { opacity: 1;
margin-top: 0; transition: .3s .15s;
}
.team-item .team-img img,
.blog-item .blog-img img { transition: .5s;
}
.team-item:hover .team-img img,
.blog-item:hover .blog-img img { transform: scale(1.15);
}
/*** Miscellaneous ***/

@media (min-width: 991.98px) {
.facts {
position: relative; margin-top: -75px; z-index: 1;
}
}

.back-to-top { position:
fixed;
display: none; right: 45px; bottom: 45px; z-index: 99;
}
.bg-header {

background: linear-gradient(rgba(9, 30, 62, .7), rgba(9, 30, 62, .7)), url(../img/web4.jpg)
center center no-repeat;
background-size: cover;
}
```

```
.link-animated a { transition: .5s;
}
.link-animated a:hover { padding-left: 10px;
}

@media (min-width: 767.98px) {
.footer-about {
margin-bottom: -75px;}

}
}
```

Chapter 9

TESTING

The purpose of testing is to discover errors. Testing is the process of trying to discover every conceivable fault or weakness in a work product. It provides a way to check the functionality of components, sub-assemblies, assemblies and/or a finished product. It is the process of exercising software with the intent of ensuring that the software system meets its requirements and user expectations and does not fail in an unacceptable manner. There are various types of tests. Each test type addresses a specific testing requirement.

Types of Testing

There are many types of testing but the common and most efficient type of testing are

1. Unit Testing
2. Integration Testing
3. System Testing
4. Acceptance Testing

9.1 : UNIT TESTING

Unit testing is testing the smallest testable unit of an application. It is done during the coding phase by the developers. Unit Testing helps in finding bugs early in the development process. Unit testing is the testing of an individual unit or group of related units. It falls under the class of white box testing. It is often done by the programmer to test that the unit he/she has implemented is producing expected output against given input.

Unit testing involves the design of test cases that validate that the internal program logic is functioning properly, and that program inputs produce valid outputs. All decision branches and internal code flow should be validated. It is the testing of individual software units of the application .it is done after the completion of an individual unit before integration.

Function	Test Description	Input	Expected Output	Pass Criteria
Login Test Case 1	Trying without Credentials	Null Values	Failed to Redirect to input page	No data
Login Test Case 2	Trying with Credentials	User Credentials	Redirect to input Page	Can enter values in input page
Predicting without entering any Data	Trying to predict data without filling anything	Null Values	Failed to predict	As there is No Input it should not predict
Predicting with Input Data	Giving URL to predict	URL	Predict whether it is safe URL or unsafe URL.	It Should accurately Predict the output.

Fig 9.1: Unit Testing

9.2 INTEGRATION TESTING

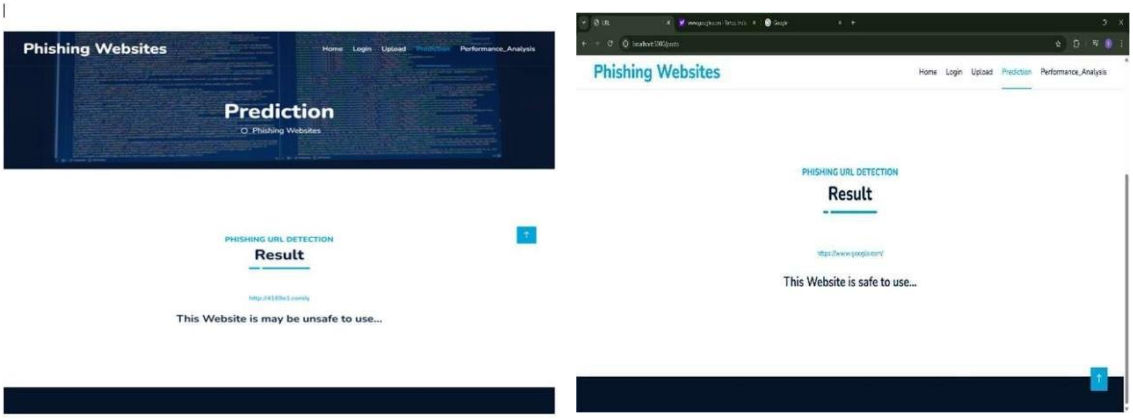
In Integration Testing different units, modules or components of a software application are tested as a combined entity. Integration tests are designed to test integrated software components to determine if they run as one program. Testing is event driven and is more concerned with the basic outcome of screens or fields. Integration tests demonstrate that although the components were individually satisfaction, as shown by successfully unit testing, the combination of components is correct and consistent. Integration testing is specifically aimed at exposing the problems that arise from the combination of components. In PWDML, the machine learning prediction system is integrated with the user interface JS code.

https://www.google.com/	legitimate
http://citeceramica.com/	phishing

Fig 9.2: Integration Testing

9.3 SYSTEM TESTING

System Testing evaluates the overall functionality and performance of a complete and fully integrated software solution. It tests if the system meets the specified requirements and if it is suitable for delivery to the end-users. The testers do not require more knowledge of programming to carry out this testing. System testing checks the interface, decision logic, control flow, recovery procedures and throughput, capacity and timing characteristics of the entire system. It will test the entire product or software so that we will easily detect the errors or defects which cannot be identified during the unit testing and integration testing.



9.4 : ACCEPTANCE TESTING

The major aim of this test is to evaluate the compliance of the system with the business requirements and assess whether it is acceptable for delivery or not. Testing here focusses on the external behavior of the system, the internal logic of the program is not emphasized. In Acceptance Testing, the system is tested for various inputs. This testing helps the project team to know the further requirements from the users directly as it involves the users for testing.

Chapter 10

SCREEN LAYOUTS

Login Page (Home Page):

The Home page of the project is the Login Page of website. On entering the username and password, the user gets login into the prediction page. After the user login into the page,we upload the required features that we need to use train the model.

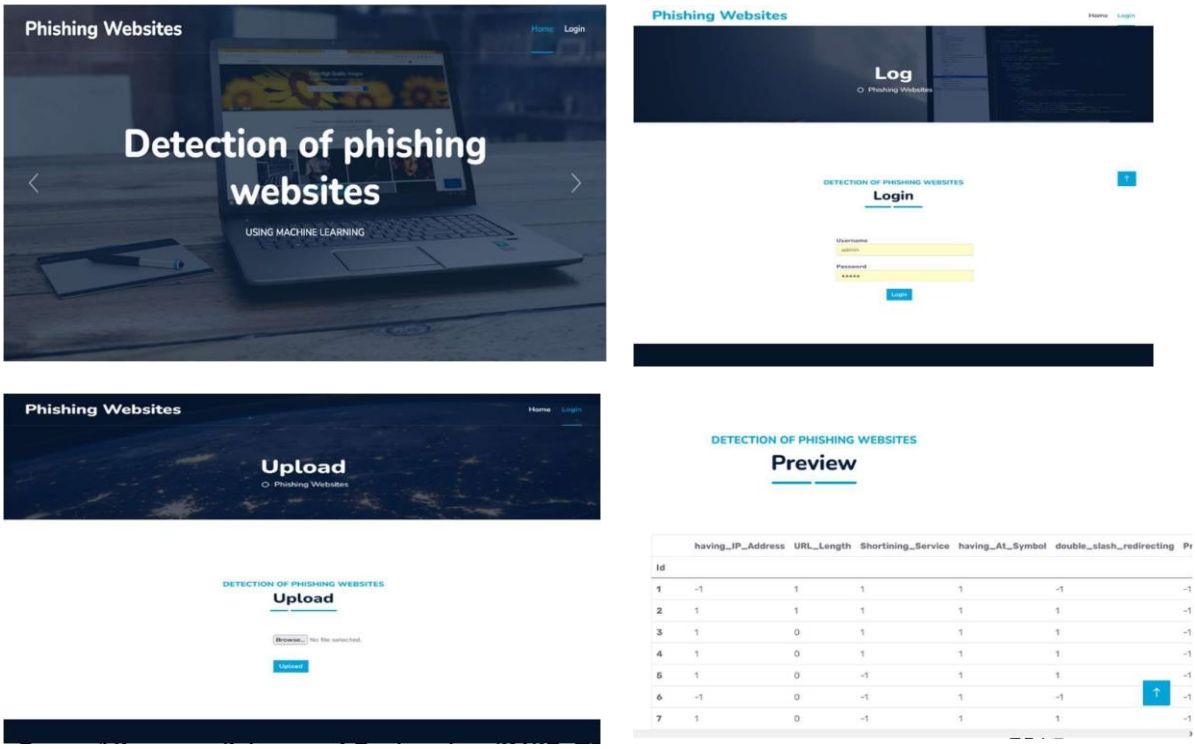


Fig 10.1: Login Page

Prediction Page:

The Prediction page of the page predict the URL, Wheather it is Legitimate(Safe) or Phishing(Unsafe).On Entering the URL that is required to predict, the Prediction System provides the results about the entered URl. If the URL is Legitimate, it gives as “Safe to use”, else it gives “Unsafe to use”

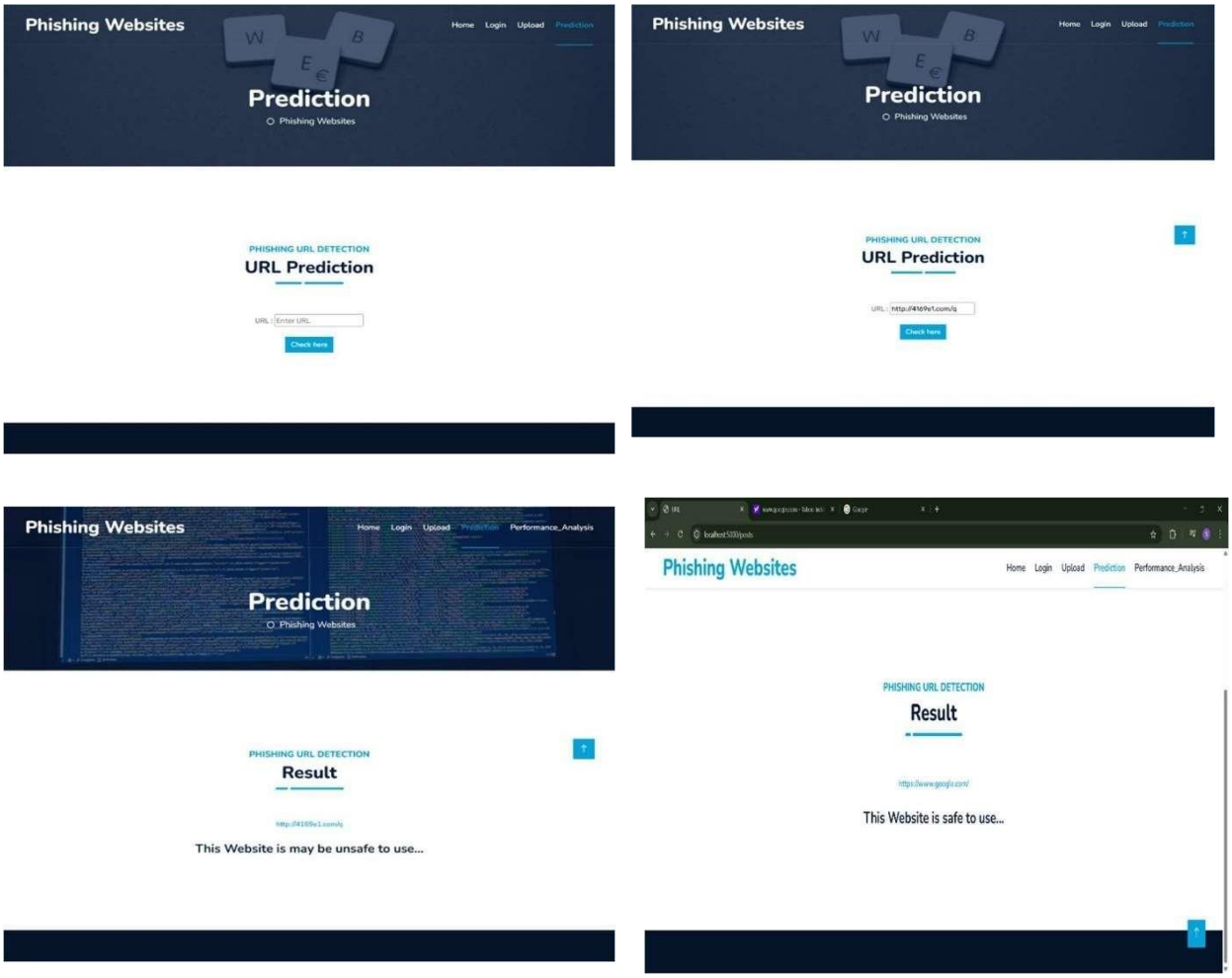


Fig 10.2: Prediction Page

Chapter 11

CONCLUSIONS

The conclusion highlights the significant benefits of a machine learning-based phishing detection system, which demonstrates immense potential in effectively addressing the growing challenge of identifying malicious URLs. By integrating and comparing multiple algorithms and selecting the most efficient one, the proposed solution ensures both high accuracy and computational efficiency. Notably, the use of the Gradient Boosting Classifier, which achieved a 97% detection accuracy with a minimal false positive rate, stands out as a powerful technique in phishing URL classification.

A key attribute of an effective anti-phishing solution is not only its precision but also its ability to predict threats in real-time or within a reasonable response window. Fast detection allows users and systems to take preventive action before any sensitive data is compromised. Therefore, the latency and processing time of the detection system become crucial in practical applications, especially for deployment on web browsers, email gateways, or even mobile devices.

Chapter 12

FUTURE ENHANCEMENTS

Although lexical features of URLs—such as length, use of special characters, and domain patterns have proven effective in achieving high accuracy in phishing detection, attackers are continuously evolving. Phishers have become adept at crafting URLs that mimic legitimate patterns, making it increasingly difficult to detect malicious links using lexical analysis alone. To counter these sophisticated evasion techniques, it is essential to combine lexical features with additional data sources such as host-based features (e.g., domain age, SSL certificate validity, DNS records) and content-based attributes (e.g., HTML structure, JavaScript behaviour, and form handling).

For future enhancements, we aim to evolve the phishing detection system into a scalable, cloud-based web service that supports real-time classification of URLs. This system will incorporate online learning techniques, enabling the model to continuously learn from newly encountered phishing patterns without requiring complete retraining. Such adaptability will significantly improve the system's resilience against zero-day attacks and evolving threats.

Chapter 13

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